Determining Transfer Function of a Motor/Flywheel System Through Experimental Methods

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***Abstract*— The objective of this lab was to experimentally obtain the transfer function for two custom-built motor/flywheel systems utilizing two experimental methods and comparing the results to theoretically obtained bode plots. Method one, the frequency domain approach, involved feeding in a series of sine waves, all of the same amplitude but varying frequencies. The system's response to each frequency was recorded and used to create a bode plot and find the transfer function of the system. Method two, the time domain approach involved feeding an impulse or step response into the system and recording how it responded to create a second transfer function. The experimental results of the two like systems were compared to the theoretical model and it was found that the frequency domain approach resulted in more accurate results for both systems. This is likely due to the fact that in the frequency approach, testing the system at one frequency at a time gives a clearer response rather than using a square wave that is inherently noisier due to its structure being made of many different frequency sine waves. This, coupled with the difficulty controlling friction in the simplified bearings used, led to a shift in magnitude between the two systems. Future reproductions of this experiment would benefit from using off-the-shelf thrust bearings to reduce system friction and higher accuracy gears to increase system response. Both of these changes should lead to less variation between systems and bode plots that are closer to the theoretical model.**

***Index Terms*—Bode Plots, Frequency Domain, Motor/Flywheel System, Time Domain, Transfer Function**

I. Introduction

Any dynamic system can be characterized by a transfer function. The transfer function of a system is an equation that represents the reaction of the system to an input and can be utilized to control a system better, with less undesirable outputs like vibrations and oscillations. Two common experimental methods can be used to find the transfer function, time domain approach and frequency domain approach. The time domain approach involves feeding an impulse or step response into your system and seeing how it responds. In an experimental electrical system, a step response might be feeding zero voltage into the system then "instantly" switching to a set voltage like 9V. This method is not always ideal as creating good, clean inputs can be difficult and the nature of the impulse can be damaging to a system due to the sudden and sometimes violent response to the input. The frequency domain approach involves feeding in a series of sine waves, all of the same amplitude but varying frequencies. The system's response to each frequency is recorded and can be used to create a bode plot and find the transfer function of the system.

For this experiment, the motor and electrical dynamics can be modeled as a lumped parameter model of a DC motor with a flywheel on the output shaft as shown in Fig.1 [1]. A lumped parameter model is a method of representing a complex system by breaking it down into a collection of idealized components like resistors and inductors.

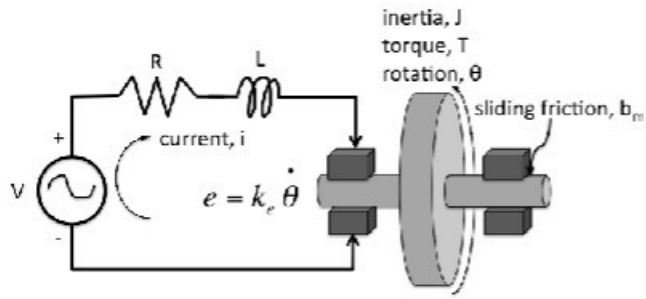

Fig. 1. A lumped capacitance model of a DC motor detailing the internal resistance (R), inductance (L), current (i), as well as the physical characteristics of the flywheel and bearing friction, J, T, θ, b.

In the lab directions, the motors dynamics are modeled as shown in (1) and the electrical systems dynamics are modeled as shown in (2) [1].

$$J\ddot{\theta} = T - b_m\dot{\theta} \tag{1}$$

$$V = iR + L\frac{di}{dt} + k_e\dot{\theta} \tag{2}$$

The Laplace transform can be taken for both systems and then combined and rearranged to find the theoretical combined transfer function for the system. Once a transfer function is found either experimentally or theoretically, a bode plot can be made to visualize the system's response in both magnitude and phase to different frequencies.

For this experiment, a motor/flywheel system was assembled as shown in [2]. The system comprised a horizontal flywheel directly connected to a DC motor via a gear set and supported by bearings. A rotational encoder was used to monitor the angle

of the flywheel and from the rate of change of the angle, the flywheel velocity could be determined after filtering the input angle. A SADI DAQ (data acquisition device) was used with a computer and external power source to control the motor and record the system's reaction. With the input voltage command and output flywheel velocity over multiple cycles, data could be analyzed, and transfer functions could be determined. The accuracy of these models was limited by the hardware shortcomings and uncontrolled environment, which could be improved upon in future experiments to generate a more accurate transfer function.

II. Procedure

The required materials for this lab were a DAQ, screwdriver, motor/flywheel system, laptop capable of running LabVIEW and a USB-C connection cord. The procedure for this lab consisted of three main parts which included constructing the motor/flywheel system, recording experimental data with LabVIEW, and analyzing the data to create and compare models of the system with bode plots and transfer functions.

Motor/Flywheel System Construction

All necessary parts and details for constructing the motor/flywheel system were provided in lab [2]. A main frame was constructed with a clamp on the base and a bearing on the top surface supporting a rotating platen. A gear was connected to the platen from below with a motor and encoder connected via a gear reduction. All wire connections for the motor, power supply, and DAQ were made to complete the assembly.

LabVIEW VI Setup

Once the motor/flywheel system was constructed, the next step was to set up the LabVIEW VI to enable data collection. A framework of the VI with instructions was provided in lab. First, a signal generator was created by adding built in sine wave and square wave functions [1]. A select operator was also added to allow the user to switch between the two functions. The proper wiring configuration was implemented following the instructions. Next, the command signal was wired to a provided output in the framework that would allow the sine and square wave inputs to be sent to the motor [1]. The instructions within the LabVIEW framework were followed to complete the remaining necessary wire and terminal connections.

Data Collection

The motor/flywheel system was plugged into a wall outlet and then wired to the EXT port on the DAQ. The DAQ was then connected to a laptop using a USB-C cord. The first step in data collection was to determine the Turn On Command to make the sine plots smoother. Frequency was set to 0.5 Hz, amplitude was set to 500 (50% motor effort), and the Turn On Command was set to 0. The sine wave option was selected and the VI was ran. A horizontal segment of data at every sign change was observed with a Turn On Command of 0. The Turn On Command was then iteratively increased until this horizontal segment dissipated into a smooth curve. This value was found to be 75 and was not adjusted for the rest of the experiment.

With the Turn On Command now determined, data collection at different frequencies could begin. The amplitude and frequency were set to 300 (30% motor effort) and 0.05 Hz, respectively, and the VI was ran with the sine wave function selected. Once a steady state response was observed, the save data button was pressed, and five cycles of data were saved before stopping the VI. The data was then saved with a .xls filename extension. This process was repeated for frequencies of 0.1, 0.2, 0.5, 1, 2, 5, 7.5, 10, 15 and 20 Hz. Once all the sine wave data was collected, the frequency was changed back to 0.2 Hz and the square wave function was selected. Five cycles of square wave data at this frequency were saved once achieving a steady state response. No additional data for square waves was needed for this experiment.

Data Analysis and Modeling

The saved data for the sine and square waves included elapsed time, command signal in bits, filtered platen rotation angle, and platen velocity in deg/s. The command signal in bits was converted to volts using (3) below where *V* and *bits* are command signal values in volts and bits, respectively.

$$V = bits * \frac{9}{1023} \tag{3}$$

Similarly, the platen velocity data in deg/s was converted to rad/s. Once these conversions were complete, a scatter plot was created that plotted both command signal in volts and platen velocity in radians per second on the vertical axis against elapsed time in seconds on the horizontal axis.

A. Frequency Domain Approach Method

The first method of modeling the system was to construct an experimental bode plot from the sine wave data at all frequencies which were converted from Hz to rad/s. First, the max input and max output had to be found for each frequency. The max input was the same for all frequencies because it was equal to the maximum amplitude of the command signal in volts which remained at 30% of motor output for the entire experiment. However, the max output at each frequency was found by taking the platen velocity value at the positive peak of the chosen cycle. The gain values for the magnitude plot were then found using (4) where *gain* is the magnitude in dB, *max output* is the peak platen velocity value in rad/s and *max input* is the peak command signal value in volts.

$$gain = 20 * \log_{10}\left(\frac{\max output}{\max input}\right) \tag{4}$$

Next, the phase values were found using (5) below where *phase* is the phase shift in degrees between the input and output signals, t_i is the time at which the command signal crossed the horizontal axis, t_o is the time the platen velocity crossed the horizontal axis and *freq* is the frequency in hertz.

$$phase = 360 * (t_i - t_o) * freq \tag{5}$$

The gain and phase values were then separately plotted on

vertical axes against frequency in radians on a log scale horizontal axis to create the experimental bode plot. A transfer function of the system could now be derived from the bode plots. The direct current (DC) gain of the system in dB, K_{DC}, was found using (6) where $gain_{0.05Hz}$ is the magnitude value in dB at a frequency of 0.05 Hz.

$$K_{DC} = 10^{\left(\frac{gain_{0.05Hz}}{20}\right)} \tag{6}$$

Next, the pole frequency, ω_o, was found by plotting two trendlines on the magnitude plot and calculating their intersection point. The first trendline was plotted for the magnitude data that remained relatively flat, which was roughly the first four to five points. The second trendline was plotted for the remaining data points that decreased in magnitude with a negative slope. The equations for these two trendlines were set equal to each other and then the pole frequency was solved. The time constant, τ, was then solved for using (7) below.

$$\tau = \frac{1}{\omega_o} \tag{7}$$

With these variables now solved for, the transfer function of the system, *TF*, was modeled using (8) where *s* is a complex frequency variable in the Laplace domain.

$$TF = \frac{K_{DC}}{\tau s + 1} \tag{8}$$

Using the DC gain and the time constant from the transfer function, a theoretical bode plot was created using MATLAB.

B. Time Domain Approach Method

The second method of modeling the system used the data from the square waves to find a transfer function. The DC gain for this method was found using (9) below where ω_i is the minimum platen velocity value in rad/s before stepping up, ω_f is the maximum platen velocity value in rad/s after stepping up for one square wave cycle and *V* is the command signal in volts.

$$K_{DC} = \frac{\omega_f - \omega_i}{2 * V} \tag{9}$$

Next, the platen velocity located at 63% the distance from ω_i to ω_o in rad/s, $\omega_{63\%}$, was found using (10) below.

$$\omega_{63\%} = 0.63 * (\omega_f - \omega_i) + \omega_i \tag{10}$$

The time constant, τ, for this model was found using (11) where $t_{63\%}$ is the time in seconds at which $\omega_{63\%}$ occurred and t_i is the time in seconds that ω_i occurred.

$$\tau = t_{63\%} - t_i \tag{11}$$

Likewise to the previous method, the transfer function was modeled with (8). Now using the square wave transfer function, another theoretical bode plot was generated using MATLAB.

III. Results

Although two motor/flywheel systems with identical components were used in this experiment, the resulting models still varied. The experimental bode plots created from the sine wave data for both systems are shown in Fig. 3 below.

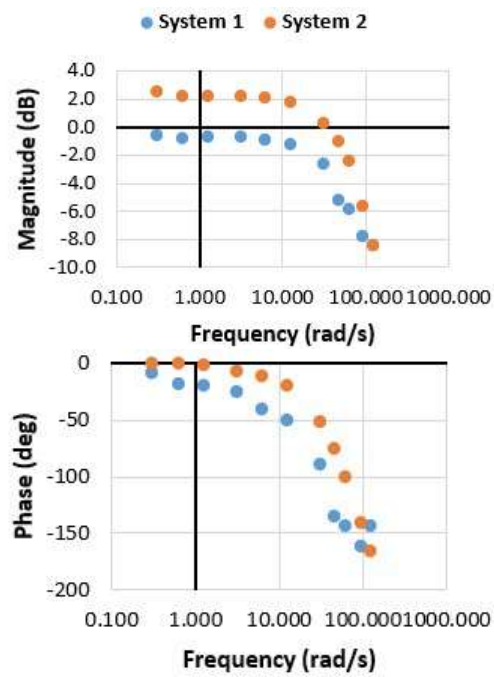

Fig. 3. Experimental bode plots for both systems.

Based off the experimental bode plots, the DC gain, K_{DC}, and time constant, τ, were calculated for both systems as shown in Table I below.

Table I
DC Gains and Time Constants from Sin Wave Data

Parameter	Value	Units
$K_{DC,1}$	0.9333	dB
$K_{DC,2}$	1.3308	dB
τ_1	0.0543	s
τ_2	0.0354	s

These values were used to create theoretical bode plots on MATLAB shown in Fig. 4 below.

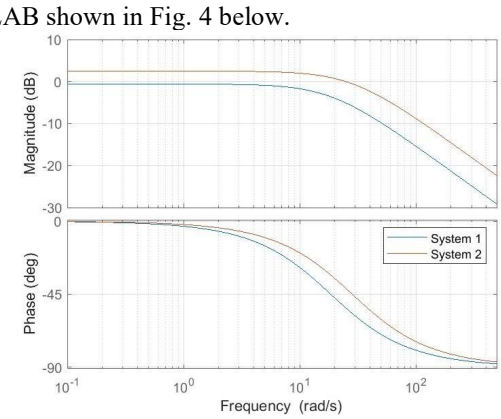

Fig. 4. Theoretical bode plots of sin wave data graphed on MATLAB for both systems.

Next, one cycle of square wave data at a frequency of 0.2 Hz for both systems was plotted as shown in Fig. 5.

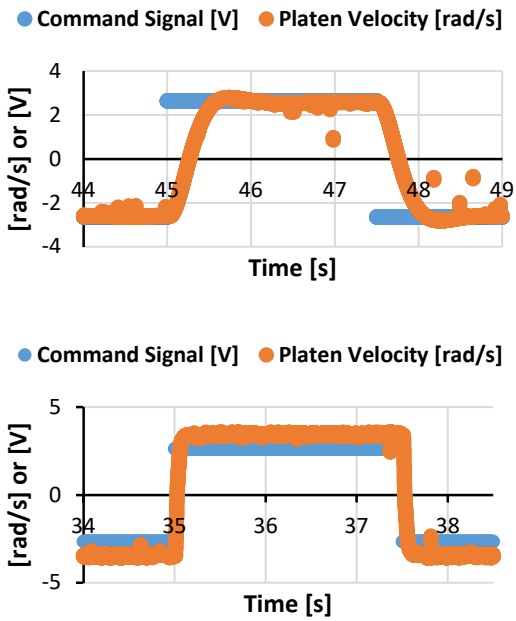

Fig. 5. Square wave data for system 1(top) and system 2 (bottom)

Based off the square wave data, the DC gain, K_{DC}, and time constant, τ, were once again calculated for both systems as shown in Table II below.

Table II
DC Gains and Time Constants from Square Wave Data

Parameter	Value	Units
$K_{DC,1}$	1.0173	dB
$K_{DC,2}$	1.3375	dB
τ_1	0.3102	s
τ_2	0.066	s

These values were used to create another set of theoretical bode plots on MATLAB shown in Fig. 6.

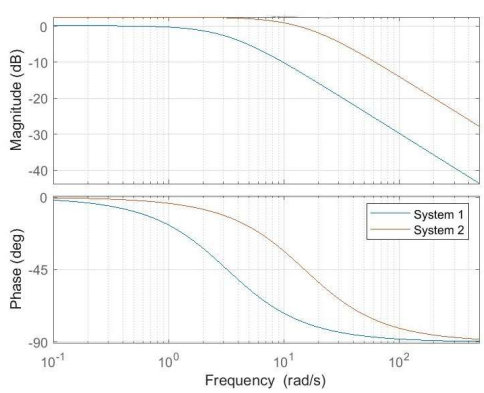

Fig. 6. Theoretical bode plots of square wave data graphed on MATLAB for both systems.

IV. Discussion

This experiment utilized two separate methods to model the dynamics of a motor/flywheel system. Method one, the frequency domain approach, relied on sine wave data to generate an experimental bode plot that a transfer function of the system was derived from. Method two, the time domain approach, relied on square wave data to determine an additional transfer function of the system. Two separate motor/flywheel systems made of the same components were tested in this experiment. After transfer functions were determined for both methods, they were used to generate theoretical bode plots on MATLAB.

The theoretical bode plots were compared to the experimental bode plots for both systems to determine if the transfer function models were derived correctly. Key points on the plots such as where the downward slope began (pole location) and the magnitude values at different frequencies were compared to determine overall accuracy. Phase plots were not compared due to experimental limitations that prevented accurate phase data. It was found that the theoretical bode plot generated from the sin wave data transfer function most closely matched the experimental bode plot for both systems. While the DC gain values derived from both methods were similar, a discrepancy in the time constant values caused the pole locations on the square wave theoretical plots to shift left further away from their location on the experimental plot.

After comparing the results of the experiment, method one was deemed the best for modeling the motor/flywheel system. A few key observations were made as to why the two systems tested differently against the theoretical bode plots. The magnitude of the two experimental systems were separated by an upwards shift on the dB scale. In a physical sense, this shift in magnitude can likely be attributed to a difference in friction in the systems as well as manufacture tolerances in the angle sensors used. The main source of friction in the systems was from the use of a simplified bearing comprised of only ball bearings and a 3D printed plastic housing with no separator, held together with self-tapping screws that could be inadvertently overtightened. Varying tolerances in the angle sensors used to calculate platen velocity, and in turn magnitude, also contributed to the shift in the bode plots between the two systems.

As for the difference in accuracy from the frequency domain and time domain approaches, the prior was found to be superior for a few reasons. By measuring the system's response at one frequency at a time, the gain and phase data was more easily extracted. A square wave, used in the time domain approach, was made up of massive amounts of summed sine waves at different frequencies which lead to a more noisy and less accurate reading. In future experiments, the platen support bearings could be replaced with off-the-shelf thrust bearings to greatly reduce friction and decrease variation between systems. Additionally, higher precision, non 3D printed gears could be used to decrease gear backlash and system sluggishness. These changes would lead to better data, allowing for a more accurate transfer function of the motor/flywheel system to be derived.

References

[1] "EML 4314C - Fall 2025 - Lab 1 Assignment" University of Florida, 2025. [Online].
[2] "Pendulum v2.3 Assembly Instructions," Out of the Box Electronics. https://www.ootbelec.com/